

NHANES Wave Profiles

A concise table report

Farnaz | October 4, 2026

Purpose

This report compares three NHANES waves using tables. It answers the requested questions about sample size, gender, ethnicity, and systolic blood pressure. The cleaned data are derived from the National Health and Nutrition Examination Survey [1].

Sample size by wave

Table 1. Number of participants in each survey wave

Wave	Participants
2015	10,175
2016	9,971
2017	9,254

The 2015 wave has the largest sample. The number of participants is slightly smaller in each later wave.

Male participants

Table 2. Male count and percentage within each wave

Wave	Male participants	Percent male
2015	5,003	49.2%
2016	4,892	49.1%
2017	4,557	49.2%

Male participants account for about half of every wave. The percentages are very similar, so no wave has a noticeably different gender balance.

Ethnicity composition

Table 3. Ethnicity count with percentage within each wave

Ethnicity	2015	2016	2017
Asian	1,074 (10.6%)	1,042 (10.5%)	1,168 (12.6%)
Black	2,267 (22.3%)	2,129 (21.4%)	2,115 (22.9%)
Mexican	1,730 (17.0%)	1,921 (19.3%)	1,367 (14.8%)
Other	470 (4.6%)	505 (5.1%)	634 (6.9%)
Other Hispanic	960 (9.4%)	1,308 (13.1%)	820 (8.9%)
White	3,674 (36.1%)	3,066 (30.7%)	3,150 (34.0%)

The composition changes across waves, and the detailed ethnicity categories make those shifts visible. Each cell reports the count followed by the percentage within its wave.

Average systolic blood pressure

Table 4. Participant-level average systolic blood pressure

Wave	Available n	Mean (SD)	Median [range]
2015	7,528	118.3 (18.1)	115.3 [64.7, 228.7]
2016	7,363	120.4 (18.4)	117.3 [74.0, 231.3]
2017	6,713	121.7 (20.3)	118.0 [72.7, 234.0]

The mean average systolic blood pressure is lowest in 2015 and highest in 2017. The available sample is smaller than the full wave sample because participants without any systolic blood pressure reading are excluded.

Conclusion

The three waves have similar gender balance but differ in sample size, ethnicity composition, and average systolic blood pressure. Counts, percentages, and summary statistics together show both the scale and the relative pattern of each comparison.

Reference

[1] National Center for Health Statistics. National Health and Nutrition Examination Survey. Centers for Disease Control and Prevention. Accessed October 4, 2026.